Report

Of

SECURE MULTI-TENANT LEGAL TECHNOLOGY PLATFORM DEVELOPMENT

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

CS2341800

AMRITESH GAUTAM

Batch: 2023–27

Section: 4CSE3

2026–27

CANDIDATE'S DECLARATION

I, AMRITESH GAUTAM, do hereby declare that the internship report titled "Secure Multi-Tenant Legal Technology Platform Development" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references used in this report have been appropriately acknowledged. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation in this regard.

Signature: 

Student Name: AMRITESH GAUTAM

Roll No.: CS2341800

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me during my industry engagement and the preparation of this report. I am thankful for the guidance, feedback and constructive discussions that helped me develop my technical understanding and contribute effectively to a real-world software product.

I am particularly grateful to **Mr. Riyad Hussain, Founder of Clearpaper**, for giving me the opportunity to work as an Independent Software Developer and for his guidance, support, and trust throughout the engagement. I am also grateful to the **Clearpaper team** for providing practical exposure to modern software engineering, product development, and collaborative problem-solving. I sincerely thank the faculty members of the **School of Computer Science and Engineering, IILM University**, for their academic guidance and support throughout this work.

I express my gratitude to my parents for their constant support and encouragement.

Signature: 

Student Name: AMRITESH GAUTAM

Roll No.: CS2341800

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate / Industry Engagement Proof	—
4.	Project Description	1
4.1	Introduction	1
4.2	Organization Profile	1
4.3	Problem Statement	2
4.4	Project Objectives	2
4.5	Scope of the Project	2
4.6	Technologies and Tools Used	3
4.7	System Architecture	3
4.8	Methodology	4
4.9	Expected Outcomes	4
4.10	Certificates of completion and other communication mail proof	5
5.	Bibliography/References	7

LIST OF TABLES

Table No.	Title
Table 4.1	Technologies and Tools Used
Table 4.2	High-Level System Architecture Components

3. INTERNSHIP COMPLETION CERTIFICATE / INDUSTRY ENGAGEMENT PROOF

A separate internship completion certificate was not issued at the time of report preparation, as the engagement was extended until October 2026 and remains ongoing. The following industry-issued extension letter is attached as official proof of the continued engagement. It confirms the role of **Independent Software Developer**, the commencement of the engagement in **January 2026**, remote part-time, and its extension until **October 2026**.

☐ Clearpaper
clearpaper.co.uk

Verification of Engagement – Amritesh Gautam

To whom it may concern,

This letter confirms that Amritesh Gautam has been engaged by Clearpaper as an Independent Software Developer since January 2026.

Amritesh works remotely on a part-time basis, supporting the design and development of the Clearpaper software platform. His engagement is currently active and remains valid until 31/10/2026. Any further extension will be subject to project requirements and mutual agreement.

Amritesh currently receives compensation of USD $500 per month.

This letter is issued at Amritesh's request for university, future placement and other official verification purposes.

Please contact me at riyad@clearpaper.co.uk if any further information or verification is required.

Riyad Hussain
Founder
riyad@clearpaper.co.uk

Date: 05/08/2026

4. PROJECT DESCRIPTION

4.1 INTRODUCTION

This report presents the work carried out during my engagement with **Clearpaper** as an **Independent Software Developer**. The engagement commenced in **January 2026** and was subsequently extended in **June 2026**, with the engagement continuing until **October 2026**. The work was performed remotely on a part-time basis in support of the development of the Clearpaper platform.

The project involved contributing to an early-stage legal technology SaaS product intended to support secure document collaboration and matter-based work for legal professionals. My responsibilities covered multiple areas of software engineering, including backend development, frontend implementation, data management, secure document infrastructure, access control, integrations and selected AI-assisted workflows.

Because the product is under active development and contains confidential implementation details, this report intentionally presents the work at a high level. Proprietary source code, internal system documentation, unreleased product details and confidential implementation specifics are not reproduced.

4.2 ORGANIZATION PROFILE

Clearpaper is an early-stage legal technology startup based in London, United Kingdom. The organization is developing a software-as-a-service platform for legal workflows, with an emphasis on secure document collaboration and organized matter workspaces. The engagement was carried out remotely within a small development team.

During the engagement, the organization operated in an early product-development stage, where engineering work involved designing, implementing and refining production-oriented software capabilities. The development environment required close collaboration, rapid iteration and attention to security, reliability and maintainability.

Organization website/domain: https://clearpaper.co.uk

4.3 PROBLEM STATEMENT

Legal professionals handle large volumes of sensitive documents and matter-related information that must be organized, accessed and shared in a controlled manner. A conventional file-sharing approach may not provide sufficient structure around matters, user roles, permissions, document versions and auditability.

The project therefore focused on contributing to a secure, multi-tenant SaaS platform that can organize legal work into isolated workspaces while supporting controlled collaboration and document management. From an engineering perspective, the challenge was to build reusable application components that maintain tenant isolation, enforce authorization rules, manage documents securely and provide a practical user experience.

4.4 PROJECT OBJECTIVES

- Contribute to the development of a secure multi-tenant SaaS platform for legal workflows.
- Implement and enhance backend services, APIs and data-management components.
- Develop frontend experiences for authenticated users and matter-based workflows.
- Support secure document storage, upload, download and version-management processes.
- Implement role- and permission-aware application behaviour across relevant workflows.
- Contribute to integrations and workflow automation required by the product.
- Develop and refine selected AI-assisted activity and time-tracking workflows.
- Improve maintainability, reliability and usability through iterative software development and testing.

4.5 SCOPE OF THE PROJECT

The scope of the engagement covered full-stack software development across several parts of the product. The work included backend API development, database-backed application logic, frontend interfaces, authentication and authorization, document-

management infrastructure, integrations and selected automation/AI-assisted workflows.

The project scope also included working with cloud-backed storage and document-processing pipelines, implementing permission-aware operations, maintaining data consistency and supporting the evolution of the product as requirements changed during development.

The scope of this academic report is intentionally limited to high-level engineering responsibilities. Internal architecture, confidential business logic, unreleased features and other proprietary information are excluded.

4.6 TECHNOLOGIES AND TOOLS USED

The following technologies and tools were used during the engagement. The selection reflects the application's full-stack architecture and supporting infrastructure.

Technology / Tool	Purpose
TypeScript / Node.js	Backend and application development
Express	REST API and server-side application layer
Next.js / React	Web application and frontend development
PostgreSQL / Prisma	Relational data storage and database access
AWS S3	Cloud-backed object storage for documents
Microsoft Graph	Integration with Microsoft services and workflows
PDF/text extraction libraries	Document text extraction and processing
Git / GitHub	Source control and collaborative development

Table 4.1 – Technologies and Tools Used

4.7 SYSTEM ARCHITECTURE

At a high level, the system follows a web-based client–server architecture. Users interact with the web application, which communicates with backend services through authenticated APIs. The backend applies business rules, authorization and data-access logic before interacting with the database and cloud storage services.

Component	High-Level Responsibility
Web Application	Provides the browser-based user interface and user-facing workflows.
Backend API	Handles authenticated requests, business rules, validation and application services.
Database	Stores structured application data such as users, workspaces, matters, permissions and document metadata.
Object Storage	Stores document files separately from application metadata and supports secure file operations.

Document Processing	Processes supported documents for text extraction and related downstream workflows.
External Integrations	Connects selected product workflows with external services such as Microsoft Graph.

Table 4.2 – High-Level System Architecture Components

The architecture description above is intentionally high-level. Detailed internal service boundaries, data models, private endpoints and implementation-specific security mechanisms are omitted to respect the confidentiality of the product.

4.8 METHODOLOGY

The engagement followed an iterative software-development approach appropriate for an early-stage product. Work was organized around individual product requirements and engineering tasks, followed by implementation, integration and verification. This allowed functionality to be developed incrementally while adapting to changing product priorities.

- Requirement understanding: clarify the intended user workflow and technical requirement before implementation.
- Design: determine the application-layer changes, data requirements and integration points needed for the task.
- Implementation: develop backend services, frontend components, database operations or integrations as required.
- Integration: connect new functionality with existing application workflows and permission rules.
- Verification: test expected behaviour, validate edge cases and resolve defects identified during development.
- Iteration: refine the implementation based on feedback and evolving product requirements.

This methodology provided practical exposure to the complete software-development lifecycle rather than isolated coding exercises.

4.9 EXPECTED OUTCOMES

- Improved understanding of production-oriented full-stack software development.

- Practical experience in designing and implementing secure multi-tenant application workflows.
- Experience with authorization, permissions and secure document-management requirements.
- Hands-on development experience with TypeScript, Node.js, Express, Next.js, React, PostgreSQL and Prisma.
- Exposure to cloud object storage, document processing and external API integrations.
- Experience contributing to AI-assisted workflows and applying engineering practices to evolving product requirements.
- Improved ability to work independently, communicate technical decisions and deliver features within a small development team.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATION MAIL PROOF

The industry extension letter attached in Chapter 3 serves as the primary engagement proof. The following email communication is attached as supporting evidence of the extension correspondence. Below is the email communications confirming the extension and continuation of my engagement with Clearpaper.

Gmail

Amritesh Gautam <amriteshx18@gmail.com>

Verification of Engagement Letter - Amritesh Gautam

1 message

Riyad Hussain <riyad@clearpaper.co.uk> Wed, Aug 5, 2026 at 5:50 AM
To: "amriteshx18@gmail.com" <amriteshx18@gmail.com>
Cc: "amritesh.gautam.cs27@iilm.edu" <amritesh.gautam.cs27@iilm.edu>

Hi Amritesh,

Please find attached your signed Verification of Engagement letter.

To continue your engagement with Clearpaper alongside your academic studies, you will need to provide a No Objection Certificate (NOC) issued by your university.

Please let me know if you have any questions.

Kind regards,
Riyad

Amritesh-Gautam-Verification-of-Engagement.pdf
44K

☐ Clearpaper
clearpaper.co.uk

Verification of Engagement – Amritesh Gautam

To whom it may concern,

This letter confirms that Amritesh Gautam has been engaged by Clearpaper as an Independent Software Developer since January 2026.

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Please contact me at riyad@clearpaper.co.uk if any further information or verification is required.

Riyad Hussain
Founder
riyad@clearpaper.co.uk

Date: 05/08/2026

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